Craniosynostosis Suture Patterns

Clinical Models for Brain Region Volumes

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Intro

Overview

- Goal: investigate deviations in volume between actual and reference heads for each region by suture type
  - Explanatory model to illustrate associations between suture type and region volume deviations
  - Predictive model to estimate future volume deviations based on suture type and previous measurements
- We use preprocessed volumetric results from CT and 3D photometry
  - Distilled to one value per region per image
  - Measures the difference in region volume between the patient's actual brain and corresponding reference head brain model
- Explanatory model shows a pattern in volume deviations associated with suture type
- Several models perform similarly for prediction

Data Description

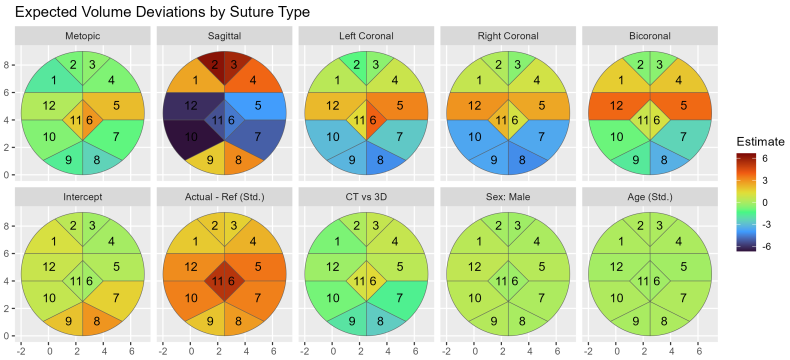
- Each row is one picture
  - The few patients who have multiple pictures have 2-3 on average, typically taken within 100 days of the first
- Demographics include **Sex, Age at image, Image Type**, Race, Ethnicity, and Genetic Diagnosis
- Columns indicate presence of suture fusions
- Each picture has associated volume measurements:
  - Actual head volume
  - Personalized normative reference head volume
  - 12 Region-specific volume deviations (Actual - Reference)
- We limit the analysis to infants less than 1 year old at the time of the picture.

Explanatory Model

Suture Fusion Associations

- Linear regression coefficients provide an estimate of the average region volume deviation for a patient in each suture group
- We also control for sex, age, image type, and overall difference between the patient's actual volume and reference head volume
- Explanatory model is a ”multivariate” regression model:
    - 12 regions' volume deviations are included as outcomes together
    - Allows for testing hypotheses about coefficients across regions
    - We can answer questions such as ”Are volume deviations in regions 1, 2, 3, and 4 related between the sagittal group and the metopic group?”
- Coefficient patterns shown on next slide:

Explanatory Coefficient Patterns



Explanatory Model Performance

Model Performance Statistics

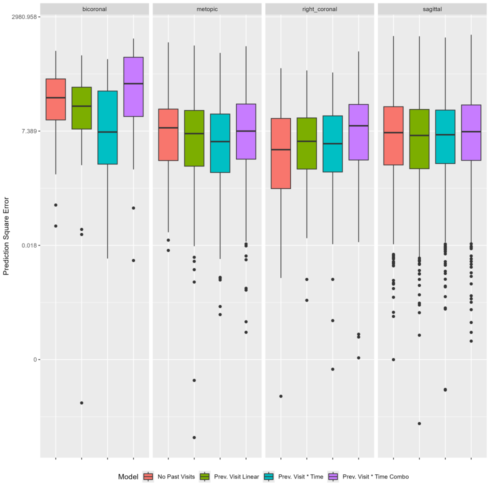
Region	RSE	R Squared	Adj. R Squared
1	2.619	0.5	0.497
2	4.297	0.447	0.443
3	3.802	0.44	0.436
4	2.847	0.561	0.558
5	3.391	0.581	0.578
6	2.892	0.785	0.784
7	4.673	0.358	0.353
8	3.901	0.528	0.525
9	3.285	0.416	0.411
10	4.971	0.373	0.369
11	2.895	0.783	0.782
12	3.113	0.612	0.609

Predictive Model

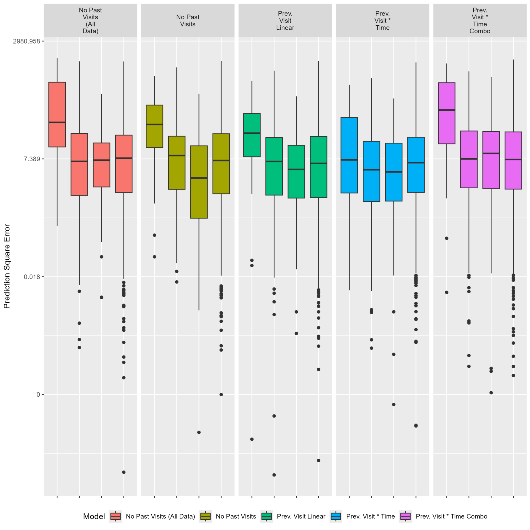
Prediction Model Description

- We want to use information from previous visits to inform predictions for future visits
- We want to check if the effect of the previous visit depends on the time since the previous picture
- In practice, we train prediction models separately for each region, including all predictors from the Explanatory Model plus:
    1. Nothing (as a baseline)
    2. Previous volume deviation as a linear term
    3. Previous value and time since previous visit as an interaction
    4. Previous value and time since previous visit interaction, using all regions together to estimate this effect
- We compare these models using **leave-one-out cross validated predictions**

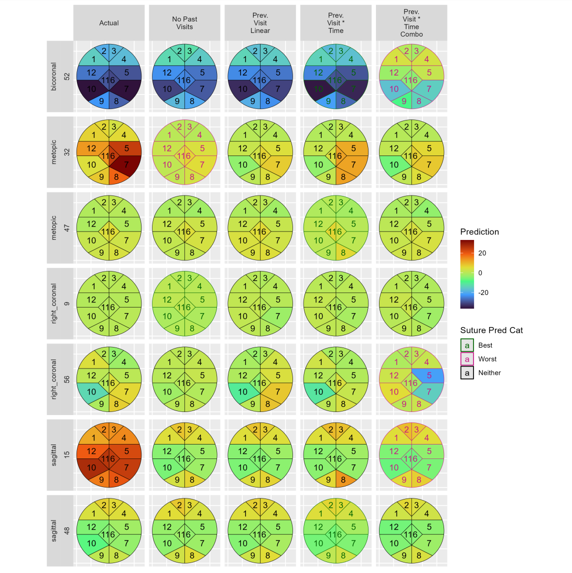
Prediction Performance i



Prediction Performance ii

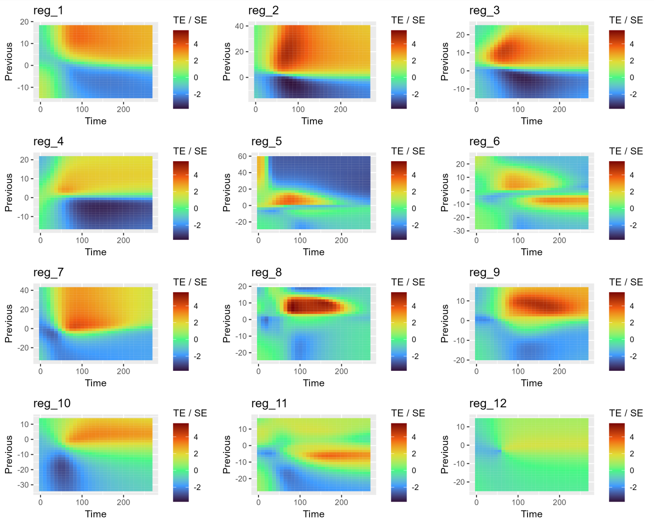


Prediction Comparison

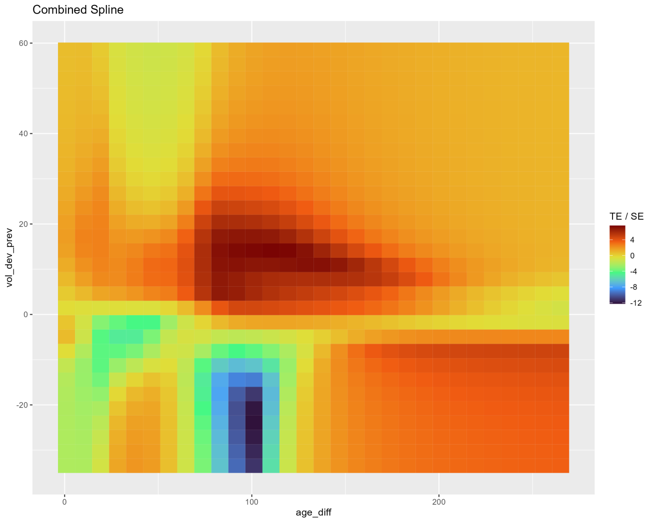


Appendix Slides

Previous Visit Effects i



Previous Visit Effects ii



Data Summaries

Suture Representation

Suture Type	Pictures	Patients	% of Patients	% of Included
no_suture	598	598	49.6%	49.6%
sagittal	371	371	30.8%	30.8%
metopic	106	106	8.8%	8.8%
bicoronal	55	55	4.6%	4.6%
right_coronal	45	45	3.7%	3.7%
left_coronal	31	31	2.6%	2.6%
Total	1206	1206	-	-

Sutures: Predictive Data

Suture Type	Pictures	Patients	% of Patients	% of Included
sagittal	65	55	62.5%	62.5%
metopic	25	20	22.7%	22.7%
right_coronal	11	7	8.0%	8.0%
bicoronal	5	5	5.7%	5.7%
left_coronal	1	1	1.1%	1.1%
Total	107	88	-	-